Module 2 Project Scenarios:

Scenario 1 (decentralized septic systems): Travel part of the human water cycle as a water molecule starting in a local drinking water source and ending in a *decentralized wastewater system*. Students will start their journey at a local drinking water source (i.e., lake), then make their way through the local drinking water system, a household (dishwashing), the household's septic system, and leave the human water cycle as part of the effluent leaching out of the lateral lines in the back yard of the house. Students will predict the expected re-entry-point into the natural water cycle and discuss any impacts the effluent may have when it re-enters the natural water cycle.

- Problem 1: You've turned on the faucet at the sink and only a dribble of water comes out.
- Problem 2: Smelly water from the septic tank is backing up into the drain in the laundry room floor.
- Problem 3: A stinky wet spot has developed in the yard between the house and the septic tank.
- Problem Solving: Base your responses to the following on what you know about the natural water cycle, have learned about the human water cycle, and any other resources to which your teacher has given you access.
    - Clarify the problem
    - Predict what may occur if the problem is not addressed
    - Predict the source(s) of this problem
    - Choose one of the sources to investigate
    - Identify and gather credible information needed to solve the problem
    - Analyze the information by comparing and contrasting alternate solutions based on several different variables
    - Evaluate the potential solutions to determine the best means to address the problem 1) *immediately* and 2) create a solution to fix the problem for the long term (may include cost/benefit analysis in this deliberation)
    - If we can, how do we prevent this problem from occurring again?

Scenario 2 (centralized sewer systems): Travel part of the human water cycle as a water molecule starting in a local drinking water source and ending in a *centralized wastewater system*. Students will start their journey at a local drinking water source (i.e., groundwater well), then make their way through the local drinking water system, a household (laundry), the sewer, a wastewater treatment facility, and leave the human water cycle as part of the effluent stream leaving the local wastewater facility. Students will predict the expected re-entry-point into the natural water cycle and discuss any impacts the effluent may have when it re-enters the natural water cycle.

- Problem 1: We are in a drought and the volume of water from the local drinking water source (shallow groundwater wells) is decreasing at such a rate that our town's current level of consumption cannot be maintained for much more than a month.
- Problem 2: A drought-buster storm system came through and dumped 10 inches of rain on our town over the course of two days.
- Problem 3: At the wastewater treatment facility, a part on one of the pumps has broken and the pump is now offline; the part is backordered for 6 months due to supply line issues.
- Problem Solving: Base your responses to the following on what you know about the natural water cycle, have learned about the human water cycle, and any other resources to which your teacher has given you access.
    - Clarify the problem
    - Predict what may occur if the problem is not addressed
    - Predict the source(s) of this problem
    - Choose one of the sources to investigate
    - Identify and gather credible information needed to solve the problem
    - Analyze the information by comparing and contrasting alternate solutions based on several different variables
    - Evaluate the potential solutions to determine the best means to address the problem 1) *immediately* and 2) create a solution to fix the problem for the long term (may include cost/benefit analysis in this deliberation)
    - If we can, how do we prevent this problem from occurring again?

Scenario 3 (plant-based agricultural systems): Travel part of the human water cycle as a water molecule starting in a local water source and ending with the field buffer strip. Students will start their journey at a local water source (i.e., river), then make their way to the field planted in some crop (e.g., corn, wheat, soybean, cotton, tobacco), exit the field as a small effluent stream from the lower corner of the field, and leave the human water cycle as the effluent stream moves into the field buffer zone that separates the field from the river. Students will predict the expected re-entry-point into the natural water cycle and discuss any impacts the effluent may have when it re-enters the natural water cycle.

- Problem 1: The farmer failed to test the soil before fertilizing and did not need to add more fertilizer.
- Problem 2: To create a private lake, one of the landowners upstream on the river has dammed up one of the larger streams that feeds the river.
- Problem 3: Recent rains have caused flooding upstream and now the crops being watered from the river appear to be stunted.

- Problem Solving: Base your responses to the following on what you know about the natural water cycle, have learned about the human water cycle, and any other resources to which your teacher has given you access.
    - Clarify the problem
    - Predict what may occur if the problem is not addressed
    - Predict the source(s) of this problem
    - Choose one of the sources to investigate
    - Identify and gather credible information needed to solve the problem
    - Analyze the information by comparing and contrasting alternate solutions based on several different variables
    - Evaluate the potential solutions to determine the best means to address the problem 1) *immediately* and 2) create a solution to fix the problem for the long term (may include cost/benefit analysis in this deliberation)
    - If we can, how do we prevent this problem from occurring again?

Scenario 4 (animal-based agricultural systems): Travel part of the human water cycle as a water molecule starting in a local water source and ending in a pasture. Students will start their journey at a local water source (i.e., groundwater well), then make their way to the water holding tank in a pasture, be consumed by a cow, and leave the human water cycle as urine released on the pasture. Students will predict the expected re-entry-point into the natural water cycle and discuss any impacts the effluent may have when it re-enters the natural water cycle.

- Problem 1: The cattle market is currently down so a local rancher has decided to get into the cattle business and turned the pasture, which is a nice, sheltered valley, into a feedlot.
- Problem 2: The price of oil has gone up and new drilling techniques make it more cost-effective to drill in new areas previously considered to be unreliable. As a result, new oilfield drilling has started in several places near the cattle pasture and the water coming out of the well now has a sheen and smell that were not there before.
- Problem 3: A rancher bought a new bull to increase fertility in his herd and had a great spring calf crop for the first time ever with 100% of his cows successfully producing a calf. It's six months later and, while it's an average autumn as far as precipitation is concerned, the well that supplies his cattle pasture appears to be running dry.
- Problem Solving: Base your responses to the following on what you know about the natural water cycle, have learned about the human water cycle, and any other resources to which your teacher has given you access.
    - Clarify the problem

    - Predict what may occur if the problem is not addressed
    - Predict the source(s) of this problem
    - Choose one of the sources to investigate
    - Identify and gather credible information needed to solve the problem
    - Analyze the information by comparing and contrasting alternate solutions based on several different variables
    - Evaluate the potential solutions to determine the best means to address the problem 1) *immediately* and 2) create a solution to fix the problem for the long term (may include cost/benefit analysis in this deliberation)
    - If we can, how do we prevent this problem from occurring again?

Scenario 5 (industrial systems): Travel part of the human water cycle as a water molecule starting in a local drinking water source and ending with the centralized wastewater system. Students will start their journey at a local drinking water source (i.e., river and groundwater wells), then make their way through the local drinking water system, an industrial site (e.g., making plastic widgets, baking bread, making batteries, materials recycling), the industrial site's water treatment facility, the sewer, a wastewater treatment facility, and leave the human water cycle as part of the effluent stream leaving the local wastewater facility. Students will predict the expected re-entry-point into the natural water cycle and discuss any impacts the effluent may have when it re-enters the natural water cycle.

- Problem 1: A local kid was fooling around with a book of matches, lit the pack, dropped it down a manhole near the industrial site, and caused an explosion that sent the manhole cover 10 feet in the air and knocked the kid to the ground.
- Problem 2: Lightning struck the electrical substation nearest the industrial site and the facility is now running on generators; due to poor maintenance, not all of the generators are fully functional.
- Problem 3: We are in a drought and the water levels in all our local drinking water sources are dropping rapidly.
- Problem Solving: Base your responses to the following on what you know about the natural water cycle, have learned about the human water cycle, and any other resources to which your teacher has given you access.
    - Clarify the problem
    - Predict what may occur if the problem is not addressed
    - Predict the source(s) of this problem
    - Choose one of the sources to investigate
    - Identify and gather credible information needed to solve the problem

- Analyze the information by comparing and contrasting alternate solutions based on several different variables
- Evaluate the potential solutions to determine the best means to address the problem 1) *immediately* and 2) create a solution to fix the problem for the long term (may include cost/benefit analysis in this deliberation)
- If we can, how do we prevent this problem from occurring again?